Relativity and Electromagnetism Homework Assignment 4

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Problem 1

The general solution for the potential is of the form:

$$V(r,\theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos\theta)$$

analyzing the problem we can immediately see that the potential must of the form:

$$V(r,\theta) = A_2 r^2 + \frac{B_2}{r^3} P_2(\cos\theta) + A_3 r^3 + \frac{B_3}{r^4} P_3(\cos\theta)$$

This is because, at the radius a sphere, the potential must be $V(a,\theta) = V_a P_2(\cos\theta)$, and at the radius b sphere, the potential must be $V(b,\theta) = V_b P_3(\cos\theta)$. there thus has to be a P_2 and P_3 term, and no other terms.

We can now use these boundary conditions to find the coefficients first at the radius a sphere, we find that:

$$\begin{cases} A_2 a^2 + \frac{B_2}{a^3} = V_a \\ A_3 a^3 + \frac{B_3}{a^4} = 0 \end{cases}$$

and at the radius b sphere, we find that:

$$\begin{cases} A_2 b^2 + \frac{B_2}{b^3} = 0 \\ A_3 b^3 + \frac{B_3}{b^4} = V_b \end{cases}$$

combining these two systems of equations we find that:

$$\begin{cases} A_2 = \frac{V_a a^3}{a^5 - b^5} \\ B_2 = \frac{V_a a^5 b^3}{b^5 - a^5} \end{cases}$$

and

$$\begin{cases} A_3 = \frac{V_b b^4}{b^7 - a^7} \\ B_3 = \frac{V_b b^4 a^7}{a^7 - b^7} \end{cases}$$

plugging these coefficients back into the general solution we find that the potential in de cavity is given by:

$$V(r,\theta) = \left(\frac{a^3}{a^5 - b^5} r^2 + \frac{a^3 b^5}{b^5 - a^5} \frac{1}{r^3} \right) V_a P_2(\cos\theta) + \left(\frac{b^4}{b^7 - a^7} r^3 + \frac{b^4 a^7}{a^7 - b^7} \frac{1}{r^4} \right) V_b P_3(\cos\theta)$$

Problem 2

for this charge configuration we find that the position vector of the charge at $z = a$, $z = 0$ and $z = -a$ are respectively given by:

$$\begin{pmatrix} 0 \\ 0 \\ a \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ -a \end{pmatrix}$$

monopole moment

The monopole moment can be calculated as:

$$q = \sum_i q_i = 2q - 2q = 0$$

We thus find that the monopole moment of this charge configuration vanishes.

dipole moment

The dipole moment can be calculated as:

$$\mathbf{p} = \sum_i q_i \mathbf{r}_i = 2q \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - q \begin{pmatrix} 0 \\ 0 \\ a \end{pmatrix} - q \begin{pmatrix} 0 \\ 0 \\ -a \end{pmatrix} = \mathbf{0}$$

We thus find that the dipole moment of this charge configuration also vanishes.

quadrupole moment

We can calculate the elements of the quadrupole moment as:

$$Q_{ij} = \frac{1}{2} \sum_k q_k (3 r_{k,i} r_{k,j} - \delta_{ij} r_k^2)$$

for the diagonal elements we find that:

$$Q_{xx} = Q_{yy} = qa^2, \quad Q_{zz} = -2qa^2$$

for the above off-diagonal elements we find that:

$$Q_{xy} = Q_{xz} = Q_{yz} = 0$$

because the quadrupole moment is symmetric, we find that the quadrupole moment of this charge configuration is given by:

$$\mathbf{Q} = \begin{pmatrix} qa^2 & 0 & 0 \\ 0 & qa^2 & 0 \\ 0 & 0 & -2qa^2 \end{pmatrix}$$

Problem 3

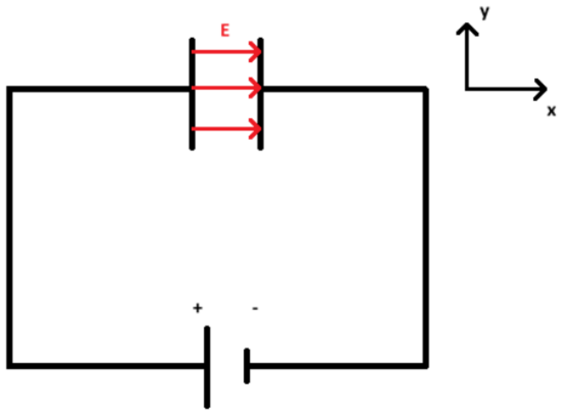

Figure 1: Circuit diagram of the problem

a.

Electric field
Per definition of the potential difference we have that:

$$V(d) - V(0) = -\int_0^d \mathbf{E} \cdot d\mathbf{l}$$

where $V(0)$ is the potential at the positive plate and $V(d)$ is the potential at the negative plate. Because the electric field is uniform between the plates, we can write that:

$$V = V(d) - V(0) = -E\int_0^d dl = -Ed$$

Than looking at the circuit, we see that the electric field is pointing in the x-direction and thus we can write that:

$$\mathbf{E} = -\frac{V}{d}\hat{\mathbf{x}}$$

Polarization
via the relation between the electric field and the polarization:

$$\mathbf{P} = \epsilon_0(\chi_1 + \chi_3 E^2)\mathbf{E}$$

we find that the polarization can be written in terms of the potential as:

$$\mathbf{P} = -\epsilon_0\left(\chi_1 + \chi_3\frac{V^2}{d^2}\right)\frac{V}{d}\hat{\mathbf{x}}$$

Displacement field
Again via the relation between the electric field, the polarization and the displacement field:

$$\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P}$$

We find that the displacement field can be written in terms of the potential as:

$$\mathbf{D} = -\epsilon_0\left(1 + \chi_1 + \chi_3\frac{V^2}{d^2}\right)\frac{V}{d}\hat{\mathbf{x}}$$

b.

Free charge

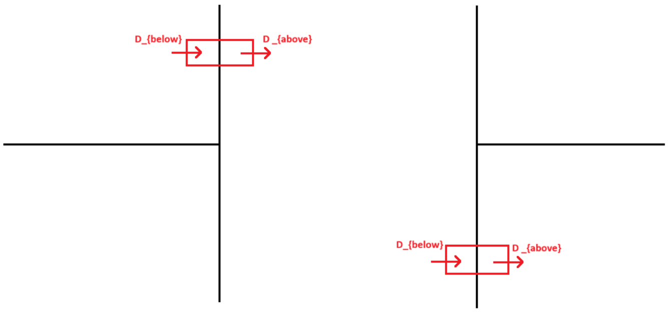

Figure 2: Circuit diagram of the problem with the free charge

In the section about boundary conditions (section 4.3.3 edition 4 of Griffiths) we find that the free charge density on the surface of the positive plate can be calculated as:

$$\sigma_f = D^{\perp}_{above} - D^{\perp}_{below}$$

for the positive plate, we only have an $D^{\perp}_{above}$ component, becous the electric field is zero outside the plates. We also have that the displacmentfield is already purpindicular to the surface of the plate, thus we can write that:

$$\sigma_f(\text{positive plate}) = D^{\perp}_{above} = -\epsilon_0 \left(1 + \chi_1 + \chi_3 \frac{V^2}{d^2}\right) \frac{V}{d}$$

for the negative plate, we only have a $D^{\perp}_{below}$ component, again because the electric field is zero outside the plates. We also have that the displacmentfield is already purpindicular to the surface of the plate, thus we can write that:

$$\sigma_f(\text{negative plate}) = -D^{\perp}_{below} = \epsilon_0 \left(1 + \chi_1 + \chi_3 \frac{V^2}{d^2}\right) \frac{V}{d}$$

Bound charge
The bounded charge surface density is defined as:

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$$

Where $\hat{\mathbf{n}}$ is the normal vector to the surface of the plate, pointing outwards. For the positive plate, we have that $\hat{\mathbf{n}} = \hat{\mathbf{x}}$, thus we can write that:

$$\sigma_b(\text{positive plate}) = \mathbf{P} \cdot \hat{\mathbf{n}} = -\epsilon_0 \left(\chi_1 + \chi_3 \frac{V^2}{d^2}\right) \frac{V}{d}$$

For the negative plate, we have that $\hat{\mathbf{n}} = -\hat{\mathbf{x}}$, thus we can write that:

$$\sigma_b(\text{negative plate}) = \mathbf{P} \cdot \hat{\mathbf{n}} = \epsilon_0 \left(\chi_1 + \chi_3 \frac{V^2}{d^2}\right) \frac{V}{d}$$